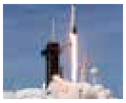
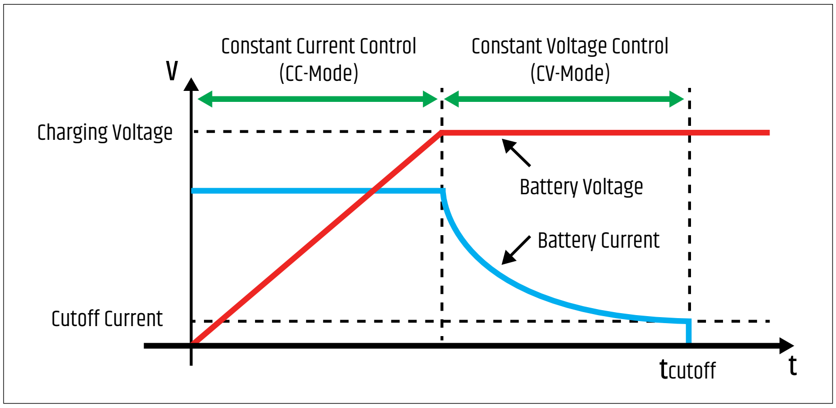
**Apple moves iPhone 13 to India**
After the iPhone SE, Apple started manufacturing the iPhone 13 in India too.
https://dgit.in/may22-29



**Private human spaceflight**
SpaceX with its Falcon 9 launched the world's first all-private human spaceflight mission to the ISS.
https://dgit.in/may22-30

The loss of mobile ions negatively impacts the maximum capacity of the battery over multiple charging cycles. This is why your 2 or 3-year old phone doesn't last quite as long as it did when it was brand new.

There are some practices to increase your battery's lifespan that you can follow. Firstly, attempt to charge your battery only up to 80 per cent and discharge only up to 20 per cent. This is because lithium-ion batteries have to work harder in the last 20 per cent of discharge and above 80 per cent of charging, so keeping the battery between those numbers can increase the overall battery lifespan. And next, try to keep your case off the phone when charging to avoid generating extra heat. This excess heat can also lead to the degradation of batteries.




Source: Research Gate

## FAST CHARGING EXPLAINED

Since there's no global standard for fast charging, manufacturers have developed their own unique methods to charge devices faster. Companies can choose to either increase the current or vary the voltage to power devices faster. A phone's charging ability is usually defined by the highest wattage supported by it. The wattage or electrical power is the rate at which electric energy is transmitted to the device. It is denoted in watts (W) or joules per second (J/s). To understand how charging works, you need to know about the terms 'watts', 'volts', and 'ampere'. Volts are a measure of voltage while amps or ampere is a measure of current. Watts is the product of volts (V) and amps (A). So, V * A = W.

Smartphones charge when a current passes through their batteries. So, greater current and higher voltages can cause them to charge at faster rates, but there's a ceiling. Battery charging is broken down into two phases – constant current and constant voltage. The voltage changes during the charging process and dictates the amount of current that can pass into a battery, as indicated by the image above.

Fast charging technologies pump as much current as possible into the battery during the constant current phase. This is before it reaches its peak voltage. Therefore, you will notice that fast chargers usually reach 50 per cent much quicker than it takes for it to go from 50 to 100.

Manufacturers are constantly developing techniques to increase the current handling capabilities of batteries to improve charging times even further. More expensive batteries even come equipped with new materials that can withstand higher currents and temperatures. Another technique used is splitting up the battery or dual-cell batteries. These batteries split the current across two batteries parallelly to give the appearance of faster charging speeds. Apart from this, algorithms are also being optimised and improved so that 'smart' chargers can better control the power delivery to a device by monitoring voltage and current.

There are various standards of fast charging for smartphones since there is no one universal standard. These include the following:

### USB Power Delivery

USB Power Delivery or USB PD is the official fast charging technology published by the USB Implementers Forum. It is one of the most commonly supported charging standards available. Many phones today sport proprietary standards of fast charging, but a majority of these phones do support USB PD as well over their USB-C port. This standard also has bi-directional power support, allowing your smartphone to reverse charge other gadgets.

### Qualcomm Quick Charge

While proprietary charging technology and USB PD standards have taken some of the limelight away from Qualcomm's Quick Charge, it still supports a range of smartphones today. The latest Quick Charge 5 is backwards compatible with all previous Quick Charge versions and USB PD. According to Qualcomm, Quick Charge or QC is also designed to be safer than the USB PD standard. This charging standard is an optional feature of the Qualcomm SoCs, so many phones powered by these chips can leverage it.

### Proprietary fast charging standards

There are multiple proprietary fast-charging protocols on the market today boasting of absurd charging speeds that one could only dream of a few years prior. Examples include VOOC and SuperVOOC from OPPO, SuperCharge from Huawei, Warp Charge from OnePlus (although newer OnePlus phones leverage SuperVOOC), Pump Express from MediaTek, DART from Realme, and others.

The future of fast charging is definitely a bright one, or should we say lightning-fast one. We have already seen companies showcase their obscenely powerful 150W and 200W chargers. In fact, Oppo announced their 240W charger at MWC 2022 that can charge a phone in just 9 minutes! We can safely assume that developments in the years to come in integrated circuitry, charge controllers, adapters and smartphones could bring charging times down even further. d